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Studierichting: Informatica
Oefeningenreeks 7C nr. 2.3

$f(x, y) = e^{(x-2y)^2} = e^{x^2-4xy+4y^2}$

Kettenregel: $(e^{g(x,y)})' = e^{g(x,y)} \cdot g'$

$\dfrac{\partial f}{\partial x} = e^{x^2-4xy+4y^2} \cdot (2x - 4y^2)$

$\dfrac{\partial f}{\partial y} = e^{x^2-4xy+4y^2} \cdot (-8xy + 16y^3)$

Productregel: $(uv)' = u'v + uv'$

$\dfrac{\partial^2 f}{\partial x^2} = e^{x^2-4xy+4y^2} \cdot (2x - 4y^2)^2 + e^{x^2-4xy+4y^2} \cdot 2$

$= e^{x^2-4xy+4y^2} \cdot (2 + (2x - 4y^2)^2)$

$\dfrac{\partial^2 f}{\partial y^2} = e^{x^2-4xy+4y^2} \cdot (-8xy+16y^3) \cdot (-8xy+16y^3) + e^{x^2-4xy+4y^2} \cdot (-8x+48y^2)$

$= e^{x^2-4xy+4y^2} \cdot [(-8xy + 16y^3)^2 + (-8x + 48y^2)]$

$= e^{x^2-4xy+4y^2} \cdot ((-8xy + 16y^3)^2 + (-8 + 48y^2))$

$\dfrac{\partial^2 f}{\partial x \partial y} = e^{x^2-4xy+4y^2} \cdot (2x - 4y^2) \cdot (-8xy + 16y^3) + e^{x^2-4xy+4y^2} \cdot (-8y)$

$\dfrac{\partial^2 f}{\partial x \partial y} = \dfrac{\partial^2 f}{\partial y \partial x} = e^{x^2-4xy+4y^2} \cdot ((2x - 4y^2) \cdot (-8xy + 16y^3) - 8y)$

References